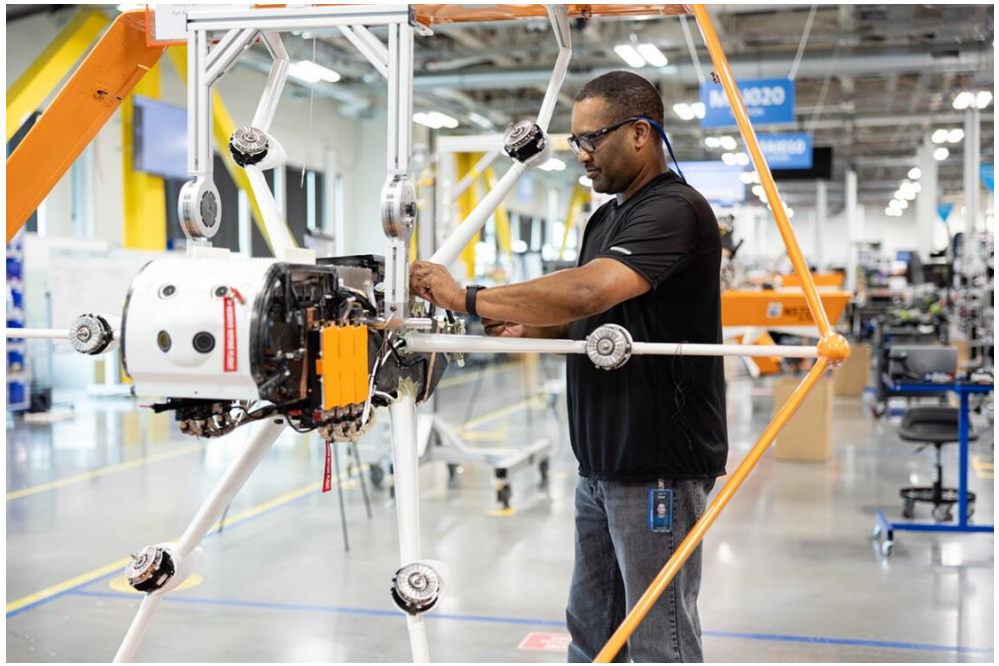
AE 510 Project: AMAZON UAV



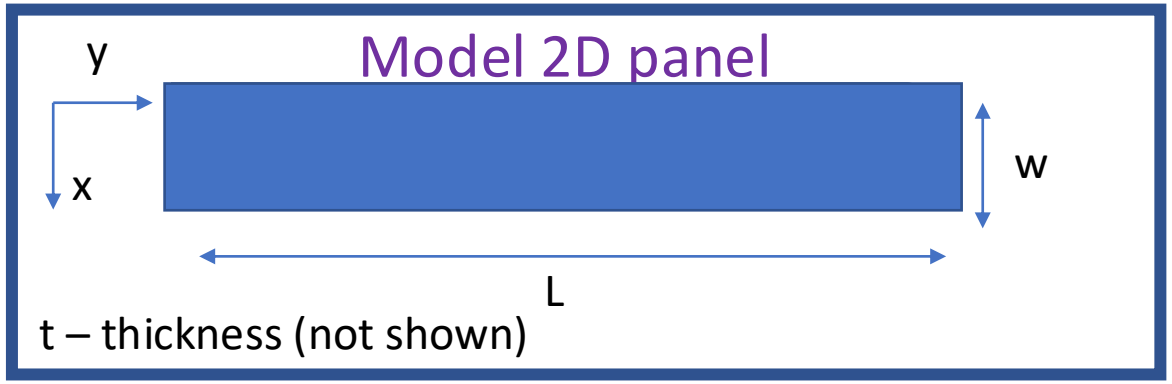
“The promise of drone delivery has often felt like science fiction. We’ve been working for almost a decade to make it a reality.

The challenge: How do you get items to customers quickly, cost-effectively, and—most importantly—safely, in less than an hour? And how do you do it in a way that can scale? It’s relatively easy to use existing technology to fly a light payload a short distance that’s within your line of sight, but it’s a very different challenge to build a network that can deliver to customers across large communities.

Our teams of hundreds of scientists, engineers, aerospace professionals, and futurists have been working hard to do just that—and later this year, Amazon customers living in Lockeford, California, will become among the first to receive Prime Air deliveries.”

Part 1: FEM ANALYSIS of AMAZON UAV

The drone weighs 90 lbs
The drone carries 5lb packages



Problem 1. Direct Method of Trusses

Problem 2. FEM of isotropic panel

Problem 3. FEM of composite panel

Submit your written report - no longer than 5 pages. (Appendix with code and other supporting documents is not included in the page count.)

Part 1: FEM ANALYSIS of AMAZON UAV

The drone weighs 90 lbs
The drone carries 5lb packages

PROBLEM 1

Model the hexagonal frame using the **Direct Method of Trusses**.

Develop 3 distinct loading cases. Apply point loads on the frame (you will need to model these and state assumptions). Calculate the deflections and stresses for each case.

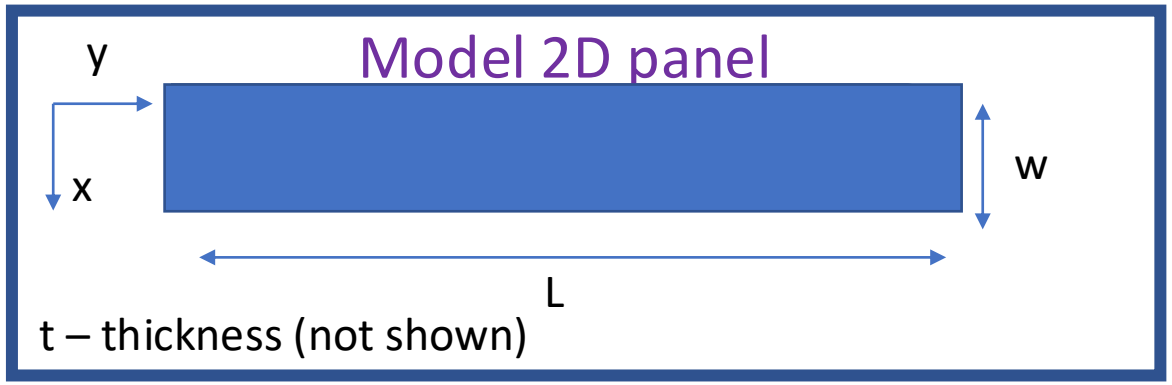
Part 1: FEM ANALYSIS of AMAZON UAV

PROBLEM 2

FEM: Model a single 2D panel (isotropic, E = 70GPa, nu = 0.3). Assume plane stress.

Generate a mesh using quad elements. Consider roughly ~ 10 elements, ~ 100 elements, ~ 1000 elements. Just use a square mesh.

Apply 3 example loading profiles (you will need to model these and state your assumptions). (For example, : i) y-axis tension on the right). Ignore aerodynamics. Determine the deflection, stress distribution, and maximum stress.



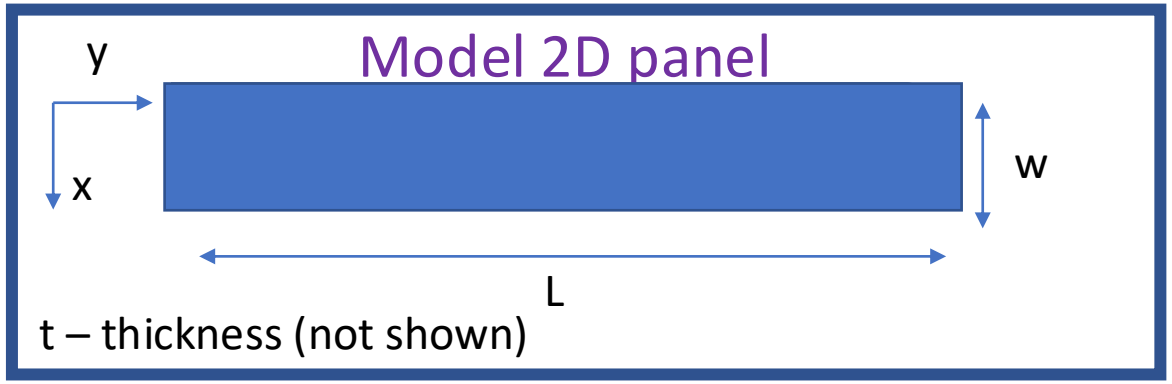
Part 1: FEM ANALYSIS of AMAZON UAV

PROBLEM 3

Model a single panel considering a CFRP composite (fixed on the left and applied tractions on the right).

See composites module for calculating orthotropic properties.

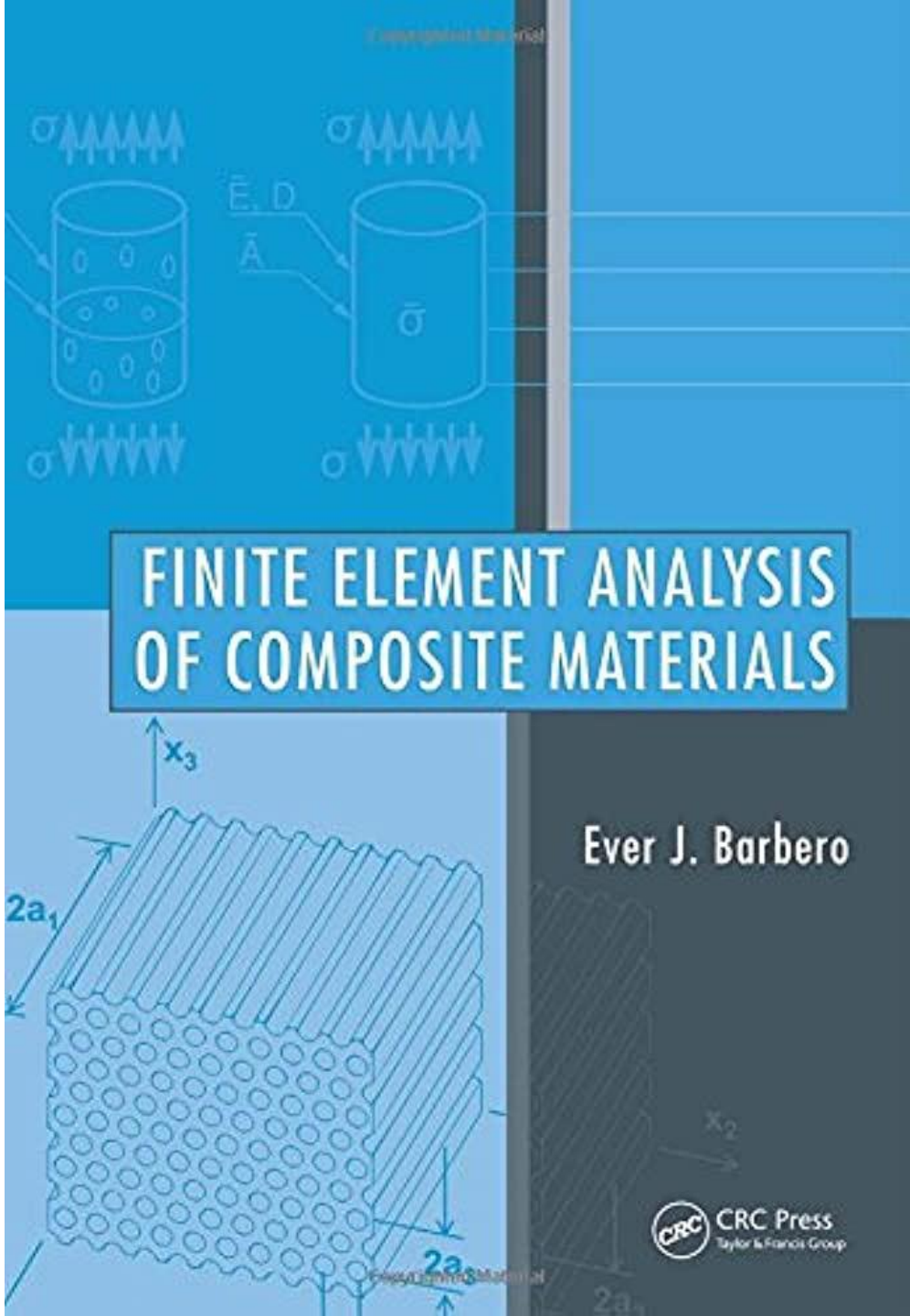
Recommend fiber orientations to minimize stress due to: i) y-axis tension on the right, and ii) arbitrary tractions on the right. Prove this is true using stress analysis for varying fiber angles.



Composite materials

- CFRP elasticity matrix for 2D plane stress case

- If you have not had 513 or 516, and want to know more, check out chapter 1 of Barbero book, and chapter 6.



Part 2: Framework to provide the context of engineering practice and societal impacts

MACROETHICS/MICROETHICS MODULE

AE 510

MicroEthics & MacroEthics

Microethics is “concerned with individuals and the internal relations of the engineering profession”

Macroethics is “concerned with the collective social responsibility of the engineering profession and societal decisions about technology”

From incandescent lightbulbs to flurorescent lightbulbs

Macroethics and Microethics

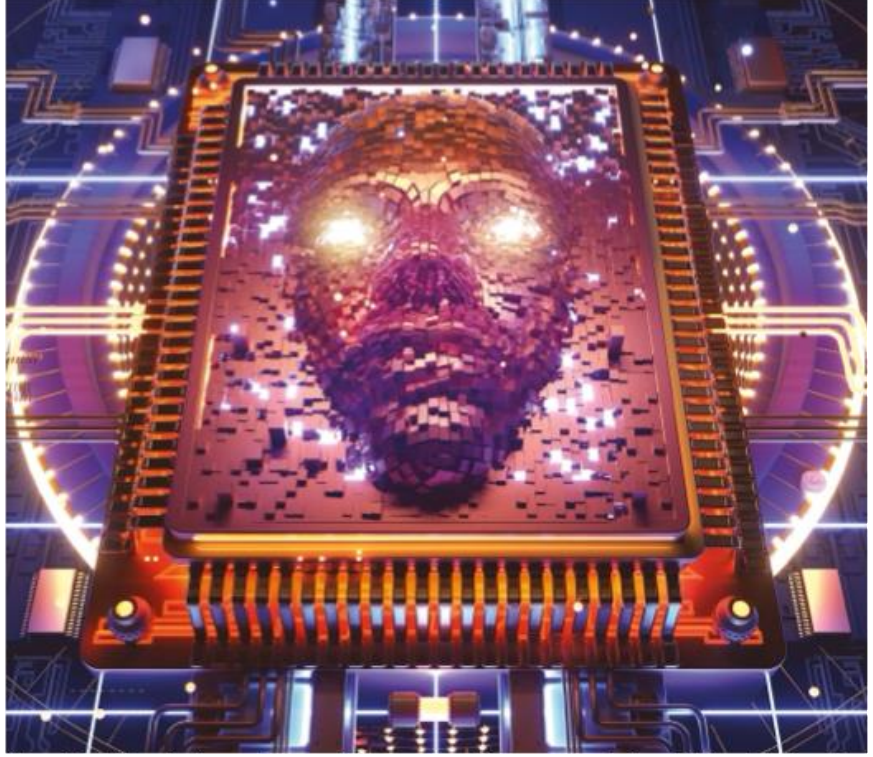
Macroethics requires the engineering community to consider the broad societal impact of science when shaping research agendas and priorities.

At the same time, microethics is needed to ensure that researchers and practitioners act in accordance with scientific and professional norms, as dictated by standards of practice, community standards of excellence, and codes of ethics.

Ethics in Science and Engineering

You must explicitly disclose the use of any genAI.

Part 2 of the project involves explicit use of AI.



Part 2a. Read the Nature article, and write a 1 page opinion/reflection.

and Stockholm conventions in 2019 following a 15-year debate. Obviously, implementing this qualified majority voting would be politically challenging and feasible only if voting was used sparingly and transparently, and without overlooking or dismissing the concerns of countries with less power.

Seize the moment

The choices made now will determine whether the health of people and the planet are safeguarded or put at further risk. We urge the INC's newly elected chair to consider implementing these reforms.

Multiple events during the past few years have undermined multilateralism. And as geopolitical priorities shift, environmental concerns are increasingly being sidelined or environmental policies weakened[10].

Against this backdrop, it is crucial that the INC process succeeds – both to address a major contributor to the interconnected planetary crises (climate change, biodiversity loss and pollution) and to restore faith in the idea that international cooperation can solve global challenges.

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1. Bergmann, M. *et al. Nature Rev. Earth Environ.* **3**, 393–337 (2022).
2. Tekman, M. B., Walther, B. A., Peter, C., Gutow, L. & Bergmann, M. *Impacts of Plastic Pollution in the Oceans on Marine Species, Biodiversity and Ecosystem* (WWF, 2022).
3. Shoett, P. *One Earth* **5**, 593–596 (2022).
4. Karali, N., Khanna, N. & Nihar, S. *Climate Impact of Primary Plastic Production* (Lawrence Berkeley National Laboratory, 2024).
5. Dauvergne, P. *et al. Mar. Policy* **181**, 106820 (2025).
6. Maes, T., Wiesnich, N., Weiland, L. & Cowan, E. *Comm. Prisms Plast.* **1**, e22 (2023).
7. MacLeod, M., Arp, H. P. H., Tekman, M. B. & Jahnke, A. *Science* **373**, 61–64 (2021).
8. UNEP. *5/14. End Plastic Pollution: Towards an International Legally Binding Instrument: Resolution/Adopted by the United Nations Environment Assembly* (UN, 2022).
9. Monclús, L. *et al. Nature* **643**, 349–355 (2025).
10. Luo, H. & Sun, Y. *Sci. Rep.* **14**, 10747 (2024).

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Does AI already have human-level intelligence? The evidence is clear

Eddy Keming Chen, Mikhail Belkin, Leon Bergen & David Danks

By any reasonable criteria, the vision of human-level machine intelligence laid out by Alan Turing in 1950 is now a reality. Eyes unclouded by dread or hype will help to prepare for what comes next.

In 1950, in a paper entitled 'Computing Machinery and Intelligence'[1], Alan Turing proposed his 'imitation game'. Now known as the Turing test, it addressed a question that seemed purely hypothetical: could machines display the kind of flexible, general cognitive competence that is characteristic of human thought, such that they could pass themselves off as humans to unaware humans?

Three-quarters of a century later, the answer looks like 'yes'. In March 2025, the large language model (LLM) GPT-4.5, developed by OpenAI in San Francisco, California, was judged by humans in a Turing test to be human 73% of the time – more often than actual humans were[2]. Moreover, readers even preferred literary texts generated by LLMs over those written by human experts[3].

This is far from all. LLMs have achieved gold-medal performance at the International Mathematical Olympiad, collaborated with leading mathematicians to prove theorems[4], generated scientific hypotheses that have been validated in experiments[5], solved problems from PhD exams, assisted professional programmers in writing code, composed poetry and much more – including chatting 24/7 with hundreds of millions of people around the world. In other words, LLMs have shown many signs of the sort of broad, flexible cognitive competence that was Turing's focus – what we now call 'general intelligence', although Turing did not use the term.

Yet many experts baulk at saying that current AI models display artificial general intelligence (AGI) – and some doubt that they ever will. A March 2025 survey by the Association for the Advancement of Artificial Intelligence in Washington DC found that 76% of leading researchers thought that scaling up current AI approaches would be 'unlikely' or 'very unlikely' to yield AGI (see go.nature.com/4smn16b).

What explains this disconnect? We suggest that the problem is part conceptual, because definitions of AGI are ambiguous and inconsistent; part emotional, because AGI raises fear of displacement and disruption; and part practical, as the term is entangled with commercial interests that can distort assessments. Precisely because AGI dominates public discourse, it is worth engaging with the concept in a more detached way: as a question about intelligence, rather than a pressing concern about social upheaval or an ever-postponed milestone in a business contract.

In writing this Comment, we approached this question from different perspectives – philosophy, machine learning, linguistics and cognitive science – and reached a consensus after extensive discussion. In what follows, we set out why we think that, once you clear away certain confusions, and strive to make fair comparisons and avoid anthropocentric biases, the conclusion is straightforward: by reasonable standards, including Turing's own, we have artificial systems that are generally intelligent. The long-standing problem of creating AGI has been solved. Recognizing this fact matters – for policy, for risk and for understanding the nature of mind and even the world itself.

Questions of definition

We assume, as we think Turing would have done, that humans have general intelligence. Some think that general intelligence does not exist at all, even in humans. Although this view is coherent and philosophically interesting, we set it aside here as being too disconnected from most AI discourse. But having made this assumption, how should we characterize general intelligence?

A common informal definition of general intelligence, and the starting point of our discussions, is a system that can do almost all cognitive tasks that a human can do[6,7].

The Atlantic My Account Give a Gift

TECHNOLOGY

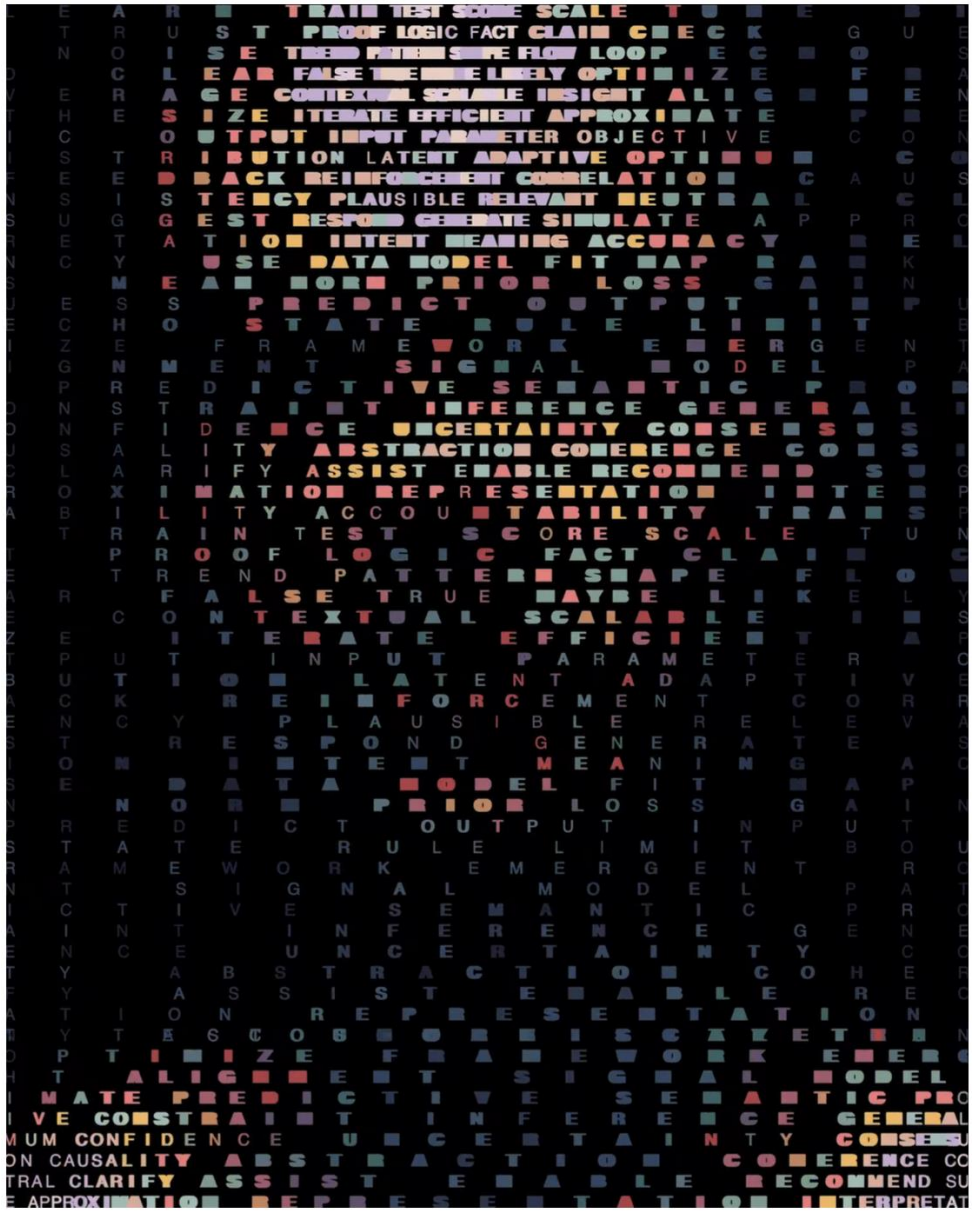
WORDS WITHOUT CONSEQUENCE

What does it mean to have speech without a speaker?

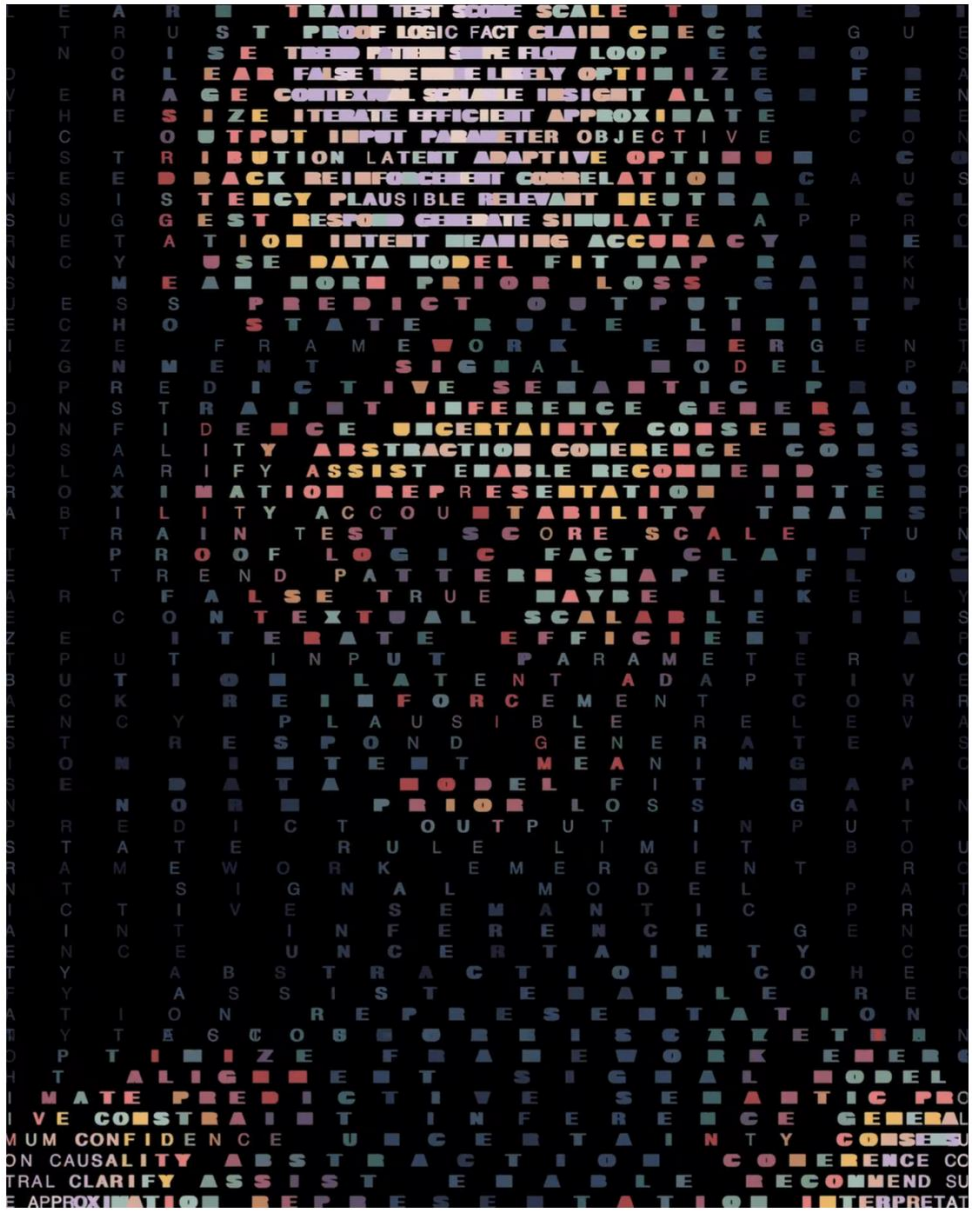
By Deb Roy

Part 2b. Read the Atlantic article, and write a 1 page reflection.

You are asked to design the structure of a UAV to avoid fracture. If the structure fails who bears responsibility?



Ask AI to do Part 1 of your project. Compare with your own work, report your findings, determine its accuracy. Provide your prompt. How do you think LLMs should be used in the engineering design and analysis process? (2 pgs)



Part 2c.
Collaborate

THE MOTIVIC CLASS OF THE SPACE OF GENUS 0 MAPS TO THE FLAG VARIETY

JIM BRYAN, BALÁZS ELEK, FREDDIE MANNERS, GEORGE SALAFATINOS, AND RAVI VAKIL

ABSTRACT. Let Fl_{n+1} be the variety of complete flags in $\mathbb{A}^{n+1}$ and let $\Omega^2_\beta(\mathrm{Fl}_{n+1})$ be the space of based maps $f : \mathbb{P}^1 \to \mathrm{Fl}_{n+1}$ in the class $f_*[\mathbb{P}^1] = \beta$. We show that under a mild positivity condition on β, the class of $\Omega^2_\beta(\mathrm{Fl}_{n+1})$ in $K_0(\mathsf{Var})$, the Grothendieck group of varieties, is given by

$$[\Omega^2_\beta(\mathrm{Fl}_{n+1})] = [\mathrm{GL}_n \times \mathbb{A}^a].$$

The proof of this result was obtained in conjunction with Google Gemini and related tools. We briefly discuss this research interaction, which may be of independent interest. However, the treatment in this paper is entirely human-authored (aside from excerpts in an appendix which are clearly marked as such).

B.1. **An early scaffold.** The following represents the first few parts of an early "scaffolding" of subproblems. These scaffolds serve (at least) two roles: (i) to explore the limits of the model's capabilities, and (ii) optionally, to provide correct solutions to include in subsequent prompts.

Prompt 1. I will ask you a question in research mathematics, so I will need you to think carefully and rigorously.

Consider $\mathbb{P}^n$, projective n-space over a field. We will work over the finite field $\mathbb{F}_q$ for now.

- I am interested in studying maps π from $\mathbb{P}^1$ to the flag variety $Fl(n+1)$ parametrizing flags in $(n+1)$-space, that send $\infty \in \mathbb{P}^1$ to a fixed flag $f \in Fl(n+1)$.
- Now each such map has some numerical invariants, corresponding to the class of $\pi_* \mathbb{P}^1$ in the Chow groups of $Fl(n+1)$.
- Precisely: there are n line bundles generating the Picard group of $Fl(n+1)$, and the degrees of the pullback of these line bundles to $\mathbb{P}^1$ are these invariants; I'll denote them d_1 through d_n.

(1) To make sure you are understanding me, could you tell me which line bundles you would choose on $Fl(n+1)$, and how you would interpret d_1 through d_n?

(2) Next, please consider the case $n = 1$. How many maps are there (over $\mathbb{F}_q$) when d_1 is one of 1, 2 or 3? Please give a rigorous proof.

(3) When you've done that, please compute the number of points (over $\mathbb{F}_q$) when $n = 2$, $d_1 = 2$ and $d_2 = 3$.

Prompting tips

- Be clear and specific in defining tasks and how you want the response structured
- Explain user, disciplinary context, how result will be used
- Use examples
- Ask to think step by step and explain reasoning
- Ask to state assumptions clearly and provide rationale explicitly
- Iterate and refine
- Ask for self-critique

The Times…

AI does and probably will continue to make mistakes. You must verify and validate responses. You must be able to judge when the result is correct and when it is not correct.

Understand that AI cannot learn for you.

Reasoning reliability and instruction-following under long context is a hard unsolved problem to date.

Ethics in Science and Engineering…

You own everything you turn in. This means you must understand and be able to explain your work. You own the consequences of the outcome of any work you produce (with or without AI).

VERIFICATION
Did we build the model correctly?
Is the implementation correct?
You check that the equations, algorithms, and code are solved properly.
Do the results converge as expected?
Do test cases with known solutions.
Challenge the implementation and results based on your expectations.

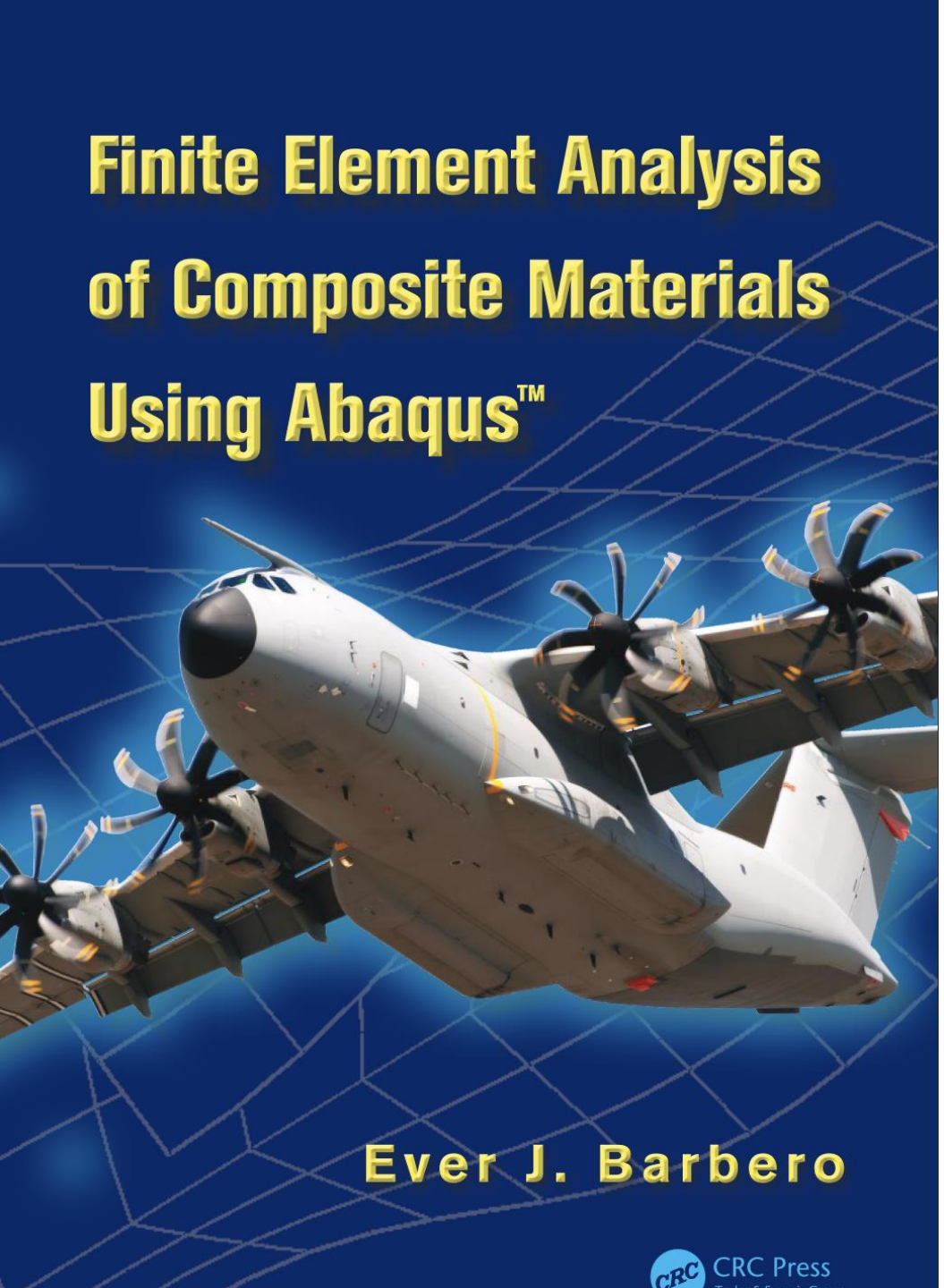
VALIDATION
Did we build the right model?
Does the model represent reality well enough?
You check that the model predictions match experimental data.
Are assumptions reasonable?
Are important physics (constraints) captured?

Critical dates

- Article discussion on April 20 (class participation 10%)
- Project due Wednesday, April 22

Grading

- Part 1 – 60% (own work)
- Part 2 – 40% - (cowork with AI)

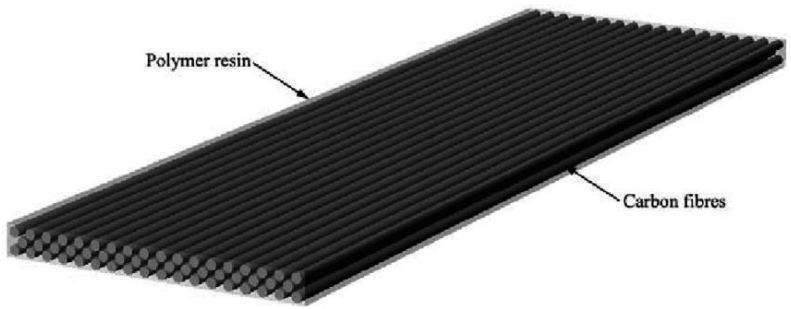


Finite element analysis of composite materials

In your class project, you replace the isotropic material with an orthotropic material. This means you replace the isotropic stiffness matrix with an orthotropic stiffness matrix for a carbon fiber reinforced polymer.

Fiber properties

- Epoxy+Carbon fiber at 60% fiber volume fraction
- E_f = 241 GPa, ν_f = 0.2, E_m = 3.12 GPa, ν_m = 0.38
- Micromechanics (chapter 6 of Barbero) will give an orthotropic matrix with equivalent stiffness



Mechanics of Orthotropic Materials

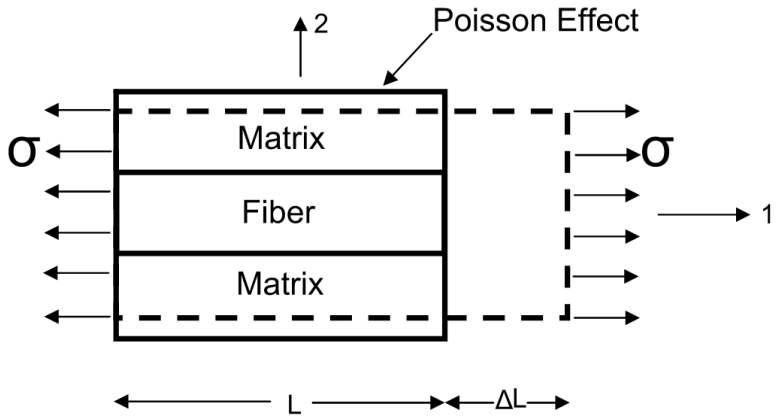
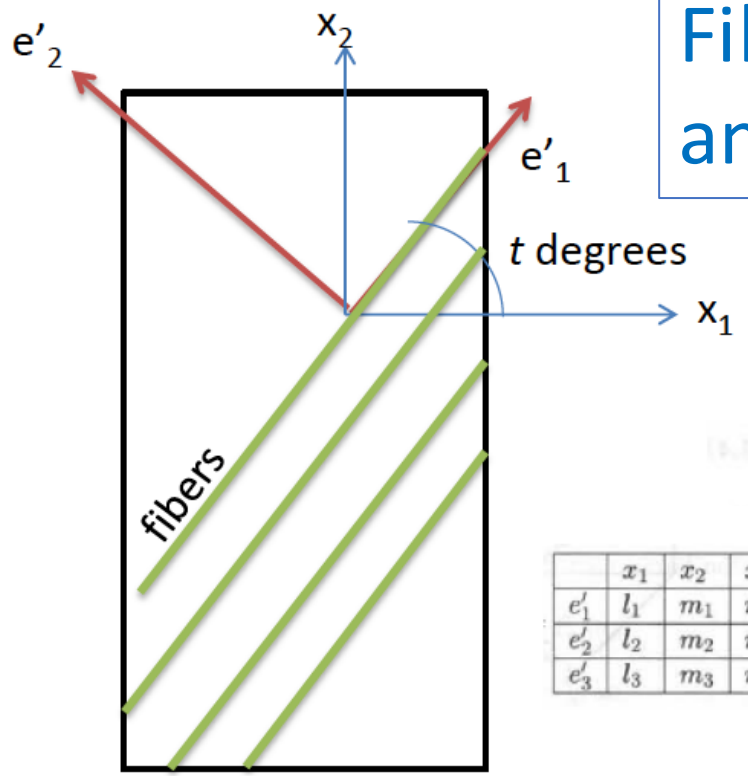


Fig. 1.13: Longitudinal loading.

These are calculated for a zero-fiber direction (of layer)

$$E_1 = 145,880 \quad G_{12} = 4,386 \quad \nu_{12} = \nu_{13} = 0.263$$
$$E_2 = 13,312 \quad G_{23} = 4,528 \quad \nu_{23} = 0.470$$

Fiber at an angle (Full 3D)



$$[S'] = \begin{bmatrix} \frac{1}{E_1} & -\frac{\nu_{12}}{E_1} & -\frac{\nu_{13}}{E_1} & 0 & 0 & 0 \\ -\frac{\nu_{12}}{E_1} & \frac{1}{E_2} & -\frac{\nu_{23}}{E_2} & 0 & 0 & 0 \\ -\frac{\nu_{13}}{E_1} & -\frac{\nu_{23}}{E_2} & \frac{1}{E_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{23}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{13}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{12}} \end{bmatrix} \tag{1.91}$$

	x_1	x_2	x_3
e'_1	l_1	m_1	n_1
e'_2	l_2	m_2	n_2
e'_3	l_3	m_3	n_3

l,m,n are cosines of the angles between axes marked

$$[C'] = [S']^{-1},$$

$$[\overline{T}] = \begin{bmatrix} l_1^2 & m_1^2 & n_1^2 & m_1 n_1 & l_1 n_1 & l_1 m_1 \\ l_2^2 & m_2^2 & n_2^2 & m_2 n_2 & l_2 n_2 & l_2 m_2 \\ l_3^2 & m_3^2 & n_3^2 & m_3 n_3 & l_3 n_3 & l_3 m_3 \\ 2l_2 l_3 & 2m_2 m_3 & 2n_2 n_3 & m_2 n_3 + n_2 m_3 & l_2 n_3 + n_2 l_3 & l_2 m_3 + m_2 l_3 \\ 2l_1 l_3 & 2m_1 m_3 & 2n_1 n_3 & m_1 n_3 + n_1 m_3 & l_1 n_3 + n_1 l_3 & l_1 m_3 + m_1 l_3 \\ 2l_1 l_2 & 2m_1 m_2 & 2n_1 n_2 & m_1 n_2 + n_1 m_2 & l_1 n_2 + n_1 l_2 & l_1 m_2 + m_1 l_2 \end{bmatrix}$$

For the above case:

l_1	m_1	n_1
l_2	m_2	n_2
l_3	m_3	n_3

=

$\cos(t)$	$\cos(90-t)$	$\cos(90)$
$\cos(90+t)$	$\cos(t)$	$\cos(90)$
$\cos(90)$	$\cos(90)$	$\cos(0)$

For negative angles (clockwise rotation of fibers), you may use the same formula with t<0. Although overall values of l,m,n will be negative compared to counterclockwise positive convention, T will still be right

$$[C] = [\overline{T}]^T [C'] [\overline{T}]$$

Plane stress reduction (full 3D to 2D)

$$\begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \gamma_4 \\ \gamma_5 \\ \gamma_6 \end{Bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} & S_{15} & S_{16} \\ S_{12} & S_{22} & S_{23} & S_{24} & S_{25} & S_{26} \\ S_{13} & S_{23} & S_{33} & S_{34} & S_{35} & S_{36} \\ S_{14} & S_{24} & S_{34} & S_{44} & S_{45} & S_{46} \\ S_{15} & S_{25} & S_{35} & S_{45} & S_{55} & S_{56} \\ S_{16} & S_{26} & S_{36} & S_{46} & S_{56} & S_{66} \end{bmatrix} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \\ \sigma_5 \\ \sigma_6 \end{Bmatrix} \begin{matrix} \\ \\ =0 \\ =0 \\ =0 \\ \\ \end{matrix}$$

For plane stress the zx,zy and zz stresses are zero

numbering

$$[\sigma] = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} \end{bmatrix} = \begin{bmatrix} \sigma_1 & \sigma_6 & \sigma_5 \\ \sigma_6 & \sigma_2 & \sigma_4 \\ \sigma_5 & \sigma_4 & \sigma_3 \end{bmatrix}$$

$$\begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ \\ \\ \\ \gamma_6 \end{Bmatrix} = \begin{bmatrix} S_{11} & S_{12} & & & & S_{16} \\ S_{12} & S_{22} & & & & S_{26} \\ & & & & & \\ & & & & & \\ & & & & & \\ S_{16} & S_{26} & & & & S_{66} \end{bmatrix} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \\ \\ \\ \sigma_6 \end{Bmatrix}$$

Remove the corresponding rows and columns

$$\begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \\ \\ \\ \sigma_6 \end{Bmatrix} = \underbrace{\begin{bmatrix} S_{11} & S_{12} & & & & S_{16} \\ S_{12} & S_{22} & & & & S_{26} \\ & & & & & \\ & & & & & \\ & & & & & \\ S_{16} & S_{26} & & & & S_{66} \end{bmatrix}^{-1}}_{\text{D matrix}} \begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ \\ \\ \\ \gamma_6 \end{Bmatrix}$$

Invert to get the D matrix

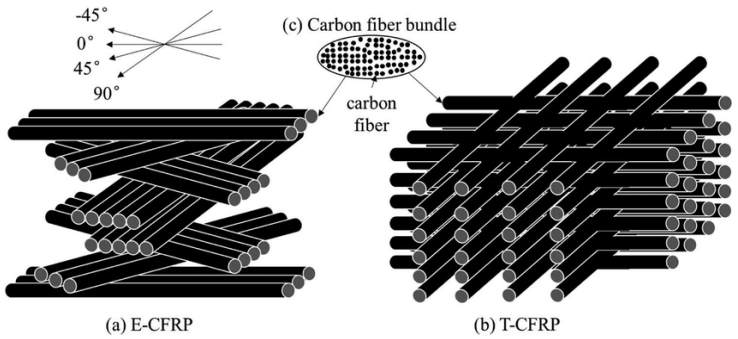
Check your composite function using textbook example

Example 1.6 *Compute the laminate properties of* $[0/90/\pm 30]_S$ *with* $t_k = 1.5$ *mm,* $E_f = 241$ *GPa,* $\nu_f = 0.2$*,* $E_m = 3.12$ *GPa,* $\nu_m = 0.38$*, fiber volume fraction* $V_f = 0.6$*, where f,m, denote fiber and matrix, respectively.*

Solution to Example 1.6 *First use periodic microstructure micromechanics (6.8) to obtain the layer properties (in MPa).*

$$E_1 = 145,880 \quad G_{12} = 4,386 \quad \nu_{12} = \nu_{13} = 0.263$$
$$E_2 = 13,312 \quad G_{23} = 4,528 \quad \nu_{23} = 0.470$$

Then, compute the compliance matrix $[S']$ *using (1.91), the rotation matrix* $[T]$ *using (1.34), the compliance* $[S]$ *in laminate coordinate system using (1.53), and the stiffness* $[C] = [S]^{-1}$ *in laminate coordinate system for each layer. Then, average them using (1.102), invert the average, and finally using (1.105) get*



The stiffness matrix $[C]$ of a balanced, symmetric laminate with N layers is built by adding the lamina matrices in laminate coordinate system multiplied by the thickness ratio t_k/t of each layer, where t is the laminate thickness and t_k denotes the thickness of the k-th layer

$$[C] = \sum_{k=1}^{N} \frac{t_k}{t}[C_k] \tag{1.102}$$

$$E_x = 78,901 \quad G_{xy} = 17,114 \quad \nu_{xy} = 0.320$$
$$E_y = 47,604 \quad G_{yz} = 4,475 \quad \nu_{yz} = 0.364$$
$$E_z = 16,023 \quad G_{xz} = 4,439 \quad \nu_{xz} = 0.280$$

$$[S] = [C]^{-1} \tag{1.103}$$

A laminate is called balanced if the total thickness of layers oriented with respect to the laminate direction at $+\theta$ and $-\theta$ are the same. Such laminate has orthotropic stiffness $[C]$ and compliance $[S]$. In terms of the apparent engineering properties of the laminate, the compliance is

Full 3D

$$[S] = \begin{bmatrix} \frac{1}{E_x} & -\frac{\nu_{yx}}{E_y} & -\frac{\nu_{zx}}{E_z} & 0 & 0 & 0 \\ -\frac{\nu_{xy}}{E_x} & \frac{1}{E_y} & -\frac{\nu_{zy}}{E_z} & 0 & 0 & 0 \\ -\frac{\nu_{xz}}{E_x} & -\frac{\nu_{yz}}{E_y} & \frac{1}{E_z} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{yz}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{xz}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{xy}} \end{bmatrix} \tag{1.104}$$

$$\begin{aligned} E_x &= 1/S_{11} & \nu_{xy} &= -S_{21}/S_{11} \\ E_y &= 1/S_{22} & \nu_{xz} &= -S_{31}/S_{11} \\ E_z &= 1/S_{33} & \nu_{yz} &= -S_{32}/S_{22} \\ G_{yz} &= 1/S_{44} \\ G_{xz} &= 1/S_{55} \\ G_{xy} &= 1/S_{66} \end{aligned} \tag{1.105}$$